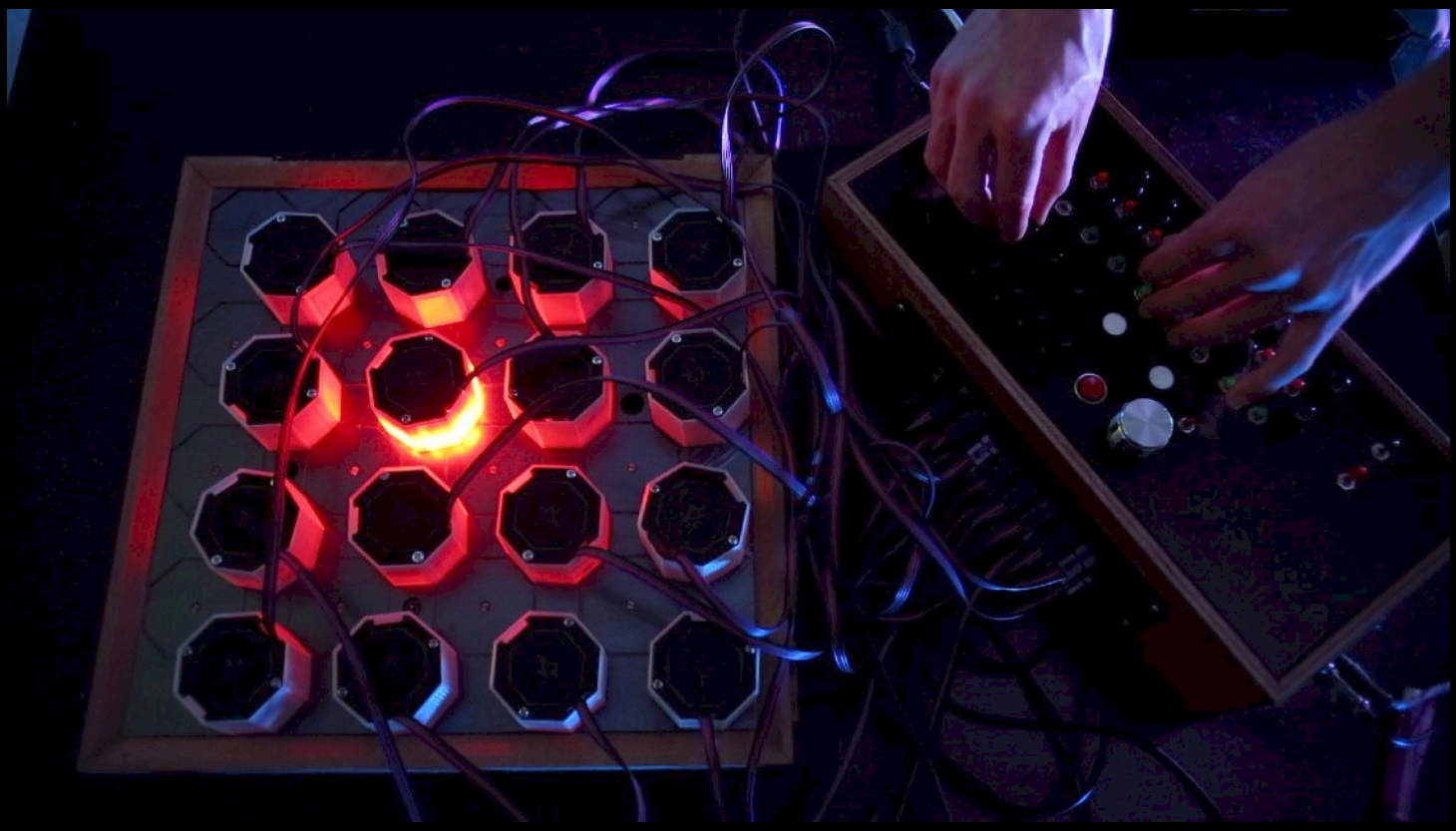
'LSe' - Project Description



Concept and Design by
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CTRL FREQ

Project supported by:

creative industries fund NL

winner @:
festival
Radical dB
4th International experimental music and sound art performance competition, Zaragoza, Spain, 2017

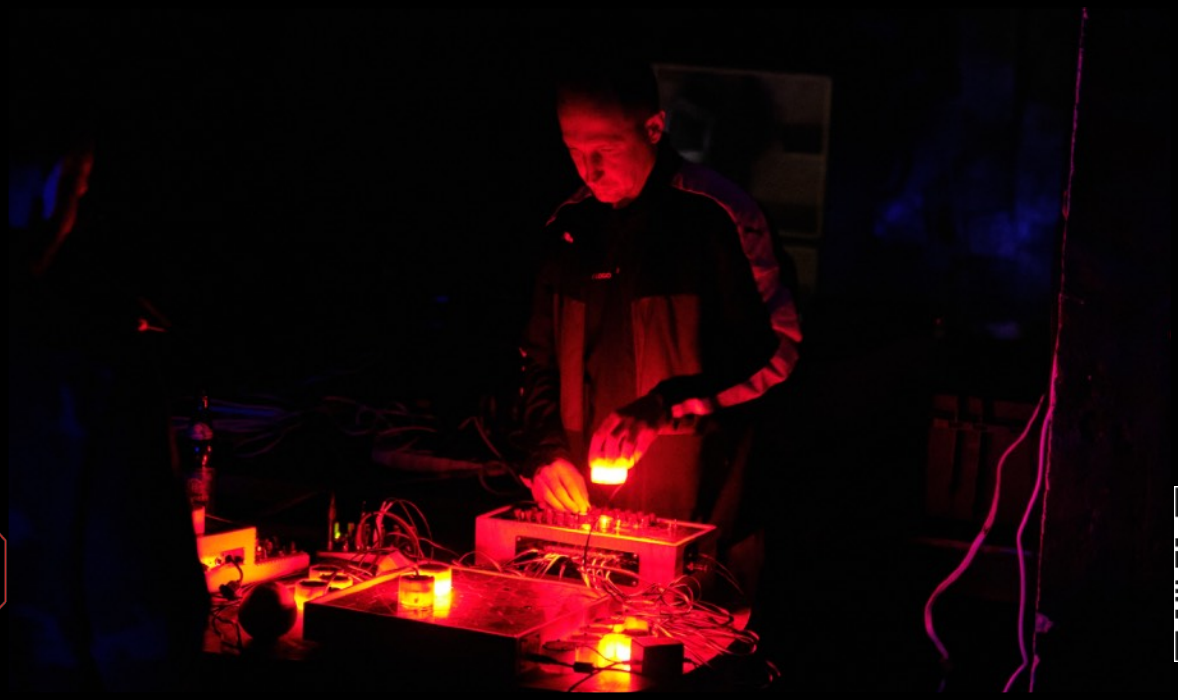
'LSe' - Live Electronic Music Performance System Controlled with Light

Introduction and general description

'***LSe***' (short for Light-Sequencer) is an innovative, award-winning instrument and performance system, specifically designed for live electronic music performance and interaction. More than a decade of research and practice has gone into this project, culminating in the recently completed third version.

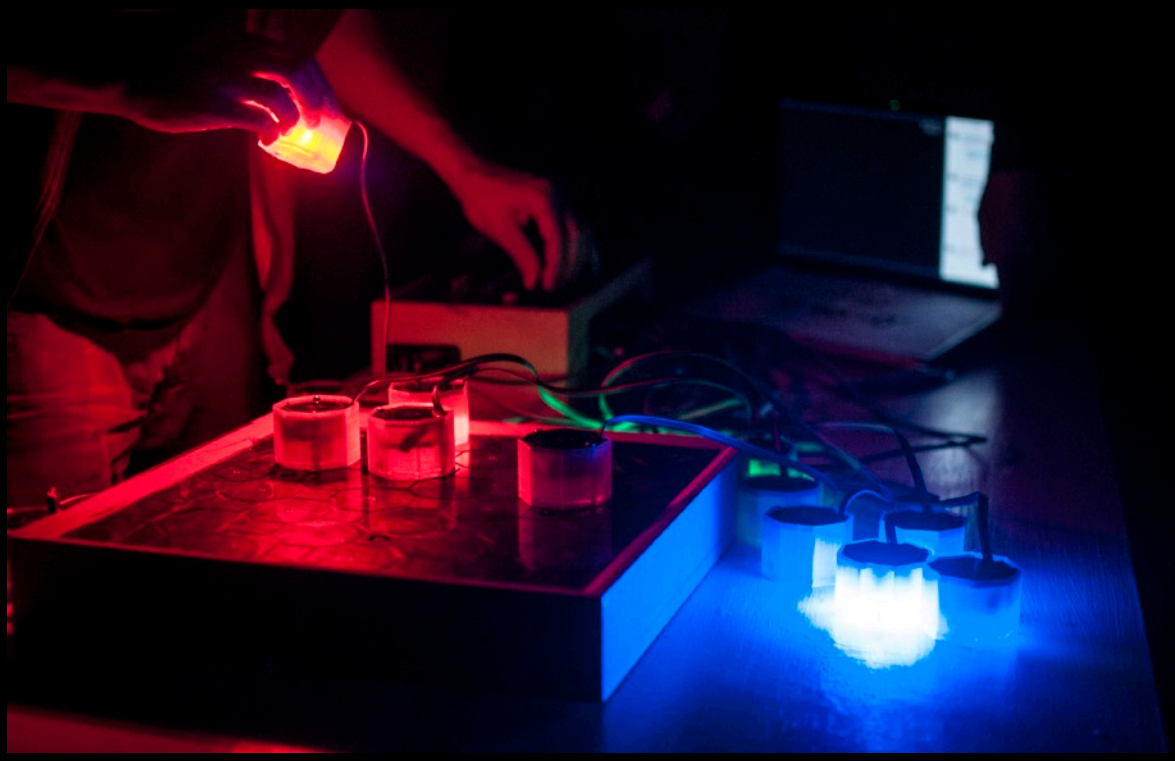
In essence, '***LSe'*** allows the creation of sequences of light that control, structure, and interact with various sound parameters. One of the main inspirations for this project was addressing the issue of 'liveness' in electronic music performances. How can the audience know if a performance is truly live? How can they relate to the musician's actions on stage? With '***LSe'***, every sound is created and manipulated live through various light sources, forging a direct connection between the visual and sonic realms and making the experience more tangible for the audience.

A picture may be worth a thousand words, but a moving image is worth a million - especially for new musical instruments and technologies that are difficult to convey through text alone. So here is the teaser of the latest version of the project.



Click on the Image to Watch the Video Teaser

‘*LSe*’ - Live Performance Examples



LSe was designed with live electronic music performance in mind. It aims to create a unique aura that can only be experienced during the event. The system works best in complete darkness, as it is light-sensitive, making it particularly camera-shy. Despite these challenges, video documentation is essential for showcasing the project. Here are some examples of ***‘LSe’*** in action:

Entire Performance (circa 18 min) at the Royal Conservatory, The Hague, NL

Entire Performance (circa 35 min) at Audio Art Festival Online Edition

Short Clip (circa 2 min)

If that isn’t enough, here is a whole YouTube playlist with more examples.

https://linktr.ee/ctrl__freq

'LSe' - Technical Details

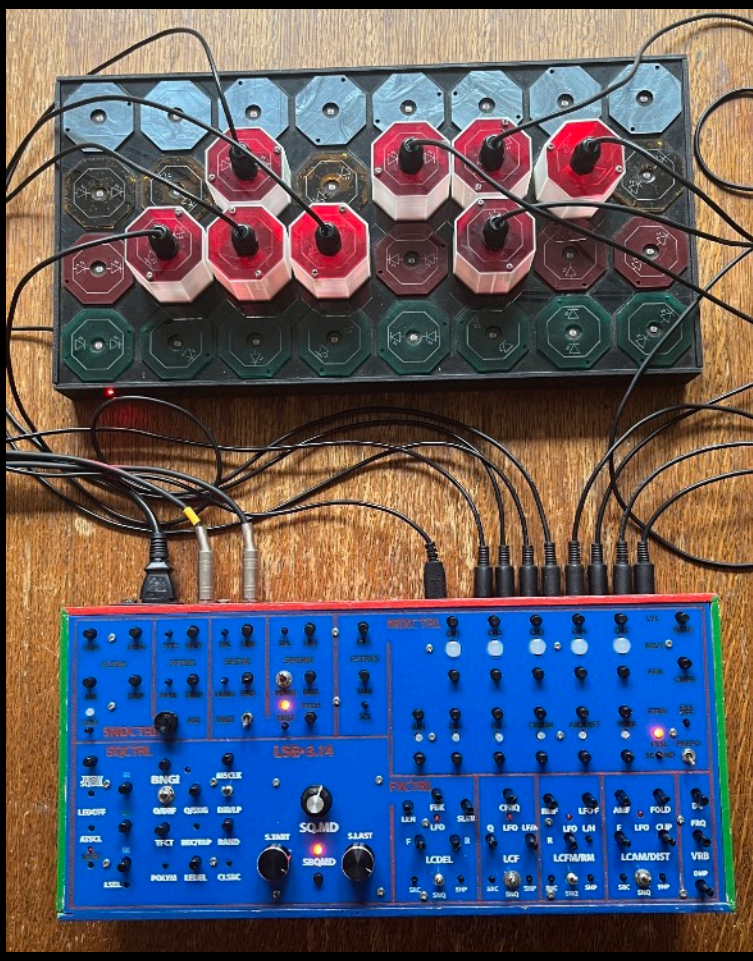
'LSe' has been my primary performance instrument since 2016, undergoing several revisions. Here are the technical details for the most recent version (3.14):

LSe version 1.0

Core Components:

- Built with Raspberry Pi, Pure Data, and Arduino
- Consists of three parts: Sensor Panel, Light Sequencer, and Control Panel
    - **Sensor Panel:** Contains 32 light-dependent resistors for sound control
    - **Light Sequencer**: Features 8 RGB Smart LEDs that can work together or independently, supporting everything from simple monophonic sequences to complex polyrhythms
    - **Control Panel:** Divided into four sections:
        - **Mix Control,**
        - **FX Control,**
        - **Sound Control,**
        - **Sequencer Control**

LSe version 3.14



Other Features:

- Full Duplex High-Quality Audio (input and output) with an Audio Injector sound card
- Ableton Link integration for WiFi or Ethernet synchronization with external gear
- Six Sequencer Modes with sub-modes (15 in total)
- Polyphony: Mono to 32 simultaneous voices
- Five Sound Generators:
    - Two synthesis-based (additive harmonic series, subtractive 3rd-octave filter)
    - Two sample-based (granular time-stretch, looper)
    - One resonant ping filter with scale selection (classic, experimental, microtuning, etc.)
- External Audio Input Channel
- Five Sound Effects Channels per generator and external audio:
    - **LCD**: Light-Controlled Delay with dedicated LFO
    - **LCF**: Light-Controlled Variable State Filter with dedicated LFO
    - **LCAM**: Light-Controlled Amplitude Modulation/Distortion with dedicated LFO
    - **LCFM**: Light-Controlled Frequency Modulation with dedicated LFO
    - **LCGV**: Light-Controlled Granular Reverb
- Light-Controlled Gate available for selected audio generators
- 'No Screen' approach, encouraging listening and playing by ear rather than relying on visual interfaces

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The life of ***'LSe'***

2005-2015	Pre-LSe era - learning electronics and programming, experimenting with light sensors, and developing early prototypes
2015-2017	Inception, funding from Creative Industries NL, Version 1 complete, multiple performances
2017-2021	Audio engine improvements, Version 2 complete, award at the Radical dB Festival, multiple performances
2021-2024	Further development, multiple concerts and presentations, Version 3 complete

Notable Appearances of ***'LSe'***

November 2024	Unsound Festival Kraków, Poland
October 2024	Sofia Experimental Festival Sofia, Bulgaria
August 2024	No Noise Festival Porto, Portugal
September 2022	Gogbot Festival, Enschede, The Netherlands
August 2022	Up To Date Festival Białystok, Poland
August 2022	Borderline Festival Wolimierz, Poland
October 2020	Café OTO Project Space London, UK
November 2019	Next Music Festival Bratislava, Slovakia
May 2017	Presentation at the NIME Conference Copenhagen, Denmark
May 2017	First Prize at the Radical dB Festival Zaragoza, Spain
November 2016	Presentation at Dutch Design Week Eindhoven, The Netherlands
October 2016	Presentation at the Amsterdam Dance Event Amsterdam, The Netherlands

Press Coverage:

Ask audio
Gear News
Matrix Synth
Audionews room
Create Digital Music

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